

Computer Science & Big Data Analytics Honours Research Report Assessment Scheme

Assessment Time

November 2025

The present scheme is adapted from the versions developed, successively, by Ian Sanders, Michael Mchunu, Hairong Bau, and Christopher Cleghorn.

Introduction

The scheme is designed to be used for assessing research reports submitted by Honours students in Computer Science and Big Data Analytics. Projects shall be classified, based on the judgment of the project supervisor, into the following categories:

- experimental computer science;
- theoretical computer science (TCS).
- philosophy of computation and artificial intelligence (PCAI).

The assessment criteria for the projects of the three types overlap, but differ. When assessing the project, only the rubrics relevant to the project type should be applied. Usually, the following areas are considered part of TCS:

- analysis of algorithms;
- computability and complexity theory;
- formal methods in computer science;
- mathematical structures in computer science.

Student Name - - - - -

Student Number - - - - -

Project Title - - - - -

Supervisor - - - - -

Mark (%) - - - - -

Component	Assessment	Mark	Weight (%)	M × W (%)
Abstract - Provides context - Explains what is done - Explains how it is done - Can be read independently of the report	Absent Very poor Poor Adequate Good Very good Excellent	0.0 1.0 1.5 2.5 3.5 4.5 5.0	× 1.0	
Introduction - Introduces general area - Introduces specific area - Introduces the work done by the author - Provides the idea of the obtained results (not necessary for TCS/PCAI projects) - Provides the right level of detail - Describes the structure of the document	Absent Very poor Poor Adequate Good Very good Excellent	0.0 1.0 1.5 2.5 3.5 4.5 5.0	× 2.0	
Problem background - Provides details of specific area - Shows familiarity with the literature - Shows understanding of the problem background - Contains relevant references - Mentions relevant ideas and concepts from the references - Relates background material to the work done by the author - The purpose of the report justified with reference to previous work	Absent Very poor Poor Adequate Good Very good Excellent	0.0 1.0 1.5 2.5 3.5 4.5 5.0	× 2.0	
Research Problem/Hypothesis/Question/Purpose - Has been clearly stated - Has been adequately justified - Is reasonable (can be solved/tested/answered/achieved)	Absent Very poor Poor Adequate Good Very good Excellent	0.0 1.0 1.5 2.5 3.5 4.5 5.0	× 1.0	
Research Methodology (not applicable to TCS/PCAI) - Is explained - Is reasonable and feasible - Is adequate to the aim of the report - Data to be collected has been identified (if applicable) - Selection of data has been justified (if applicable)	Absent Very poor Poor Adequate Good Very good Excellent	0.0 1.0 1.5 2.5 3.5 4.5 5.0	× 3.0	
Results (not applicable to TCS/PCAI) - Presented clearly - Enough empirical results have been obtained to answer questions - Author shows clear understanding of results - Results are clearly related to research question/aim - Limitation(s) of results have been discussed - Work is placed in the context of other research - Code visibility or availability (if applicable)	Absent Very poor Poor Adequate Good Very good Excellent	0.0 1.0 1.5 2.5 3.5 4.5 5.0	× 6.0	

Component	Assessment	Allocation	Rating (%)	Awarded (%)
Evidence of mastery of theoretical material (TCS only)				
• The report shows understanding of the topic	Absent	0.0	× 9.0	
• The report contains extensions of the work presented in the literature	Very poor	1.0		
• The report contains variations on the work presented in the literature	Poor	1.5		
• The topic is presented clearly and concisely	Adequate	2.5		
• The report contains evidence of independent work done by the author	Good	3.5		
• The report presents uniformly material from multiple sources	Very good	4.5		
	Excellent	5.0		
Evidence of mastery of philosophical material (PCAI only)				
• The report puts forward a clear central claim (or set of claims)	Absent	0.0	× 9.0	
• The report puts forward a cogent and rigorous argument in support of the claim(s)	Very poor	1.0		
• The argument includes possible lines of objection and their responses	Poor	1.5		
• The report shows an understanding of the debate within which the argument is situated	Adequate	2.5		
• The report shows an understanding of the logical relations relevant to the above	Good	3.5		
• The report shows an understanding of the logical relations relevant to the above	Very good	4.5		
• The argument shows originality	Excellent	5.0		
Conclusion				
• Provides a clear summary of the major points of the report	Absent	0.0	× 1.0	
• Highlights the most important results (if applicable, not necessary for TCS)	Very poor	1.0		
• Makes clear the author's contribution	Poor	1.5		
• Mentions ideas for future work	Adequate	2.5		
	Good	3.5		
	Very good	4.5		
	Excellent	5.0		
Presentation				
• The layout and typography of the document	Absent	0.0	× 3.0	
• The figures are large enough, and the pages are numbered	Very poor	1.0		
• The graphs are clear, well labelled, and properly captioned	Poor	1.5		
• The tables are well laid out and properly captioned	Adequate	2.5		
• Document structure	Good	3.5		
• Spelling, grammar, and punctuation	Very good	4.5		
• Document structure (section breakdown, etc.)	Excellent	5.0		
References				
• Are complete	Absent	0.0	× 1.0	
• Are accurate	Very poor	1.0		
• Are cited in the approved style	Poor	1.5		
	Adequate	2.5		
	Good	3.5		
	Very good	4.5		
	Excellent	5.0		

TOTAL MARK _ _ _ _ _ %